

A Soliton Cellular Automaton

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A cellular automaton (CA) of filter automata type is proposed. Any state of the CA consists only of solitary wave solutions. It is shown that the solitary waves interact with one another preserving their identities during a time evolution. It is also shown that the CA has infinitely many conserved quantities. Hence, this CA may be considered to be one of the simplest soliton systems.

[cellular automaton, filter automaton, soliton, solitary wave]

Cellular automata (CA's) serve as simple models of dynamical systems. Many types of CA's have been proposed in order to simulate various physical, chemical or biological phenomena.¹⁾ Strings of bits in the one-dimensional CA often behave like waves during a time evolution. In some cases, they interact with one another preserving their identities, which means that the waves are considered to be solitons. In 1986, Park, Steiglitz and Thurston²⁾ proposed a new kind of CA called filter automaton, and observed that soliton collisions are quite common in the CA. Filter automaton is a CA, the site values of which depend not only on ones at previous time but also on simultaneous ones. An alternative formulation of this filter automaton has been given recently, and many analytical features including stability, soliton interaction and periodic particles have been studied.^{3,4)}

In this paper, we propose a CA of the filter automaton type and investigate various properties of the system. We shall show that any state of the CA consists only of solitary waves and all the waves interact in the same manner as solitons. Moreover, we shall show that the CA possesses infinitely many conserved quantities. Hence, this CA may be considered to be one of the simplest analog of soliton equations.

First, we define a rule of time evolution of

this CA. The CA is 1 + 1 dimensional and two valued. Let u_j^t be 1 or 0, which denotes the value at integer time t and at integer space site j . Ranges of t and j are both from $-\infty$ to $+\infty$ and it is assumed that $u_j^t = 0$ for j far enough to the left and to the right. The rule of the time evolution is given by

$$u_j^{t+1} = \begin{cases} 1 & \text{if } u_j^t = 0 \text{ and } \sum_{i=-\infty}^{j-1} u_i^t > \sum_{i=-\infty}^{j-1} u_i^{t+1} \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

The value u_j^{t+1} at the site j depends on u_i^t and u_i^{t+1} with $i < j$. It is to be noted that the rule is spatially nonlocal. Therefore, this CA is an extension of filter automata. We show an example of time evolution of u_j^0 in Fig. 1. The CA is reversible in time and the rule to calculate a state at t from that at $t+1$ is given by

$$u_j^t = \begin{cases} 1 & \text{if } u_j^{t+1} = 0 \text{ and } \sum_{i=j+1}^{\infty} u_i^{t+1} > \sum_{i=j+1}^{\infty} u_i^t \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

If we define S_j^t by

$$S_j^t = \sum_{i=-\infty}^j u_i^t, \quad (3)$$

eq. (1) reduces to

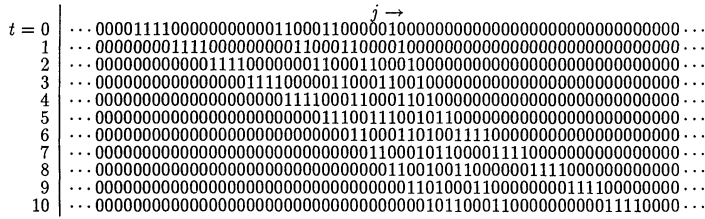


Fig. 1. Example of time evolution according to eq. (1).

$$S_j^{t+1} - S_{j-1}^{t+1} = \frac{1}{2\delta} (1 - S_j^t + S_{j-1}^t)(S_{j-1}^t - S_{j-1}^{t+1} + \delta - |S_{j-1}^t - S_{j-1}^{t+1} - \delta|), \quad (4)$$

where δ is a constant and $0 < \delta \leq 1$. Equation (4) means that eq. (1) may reduce to a nonlinear finite-difference equation in which a dependent variable S_j^t is nonnegative integer and the evolution rule of which is spatially local.

If a state at t is related to a state at $t+1$ as

$$u_j^{t+1} = u_{j+k}^t, \quad (5)$$

for all j and for a constant integer k , then we have

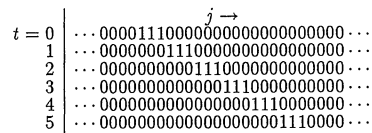
$$u_j^{t+l} = u_{j+kl}^t, \quad (6)$$

for all j and for an arbitrary integer l . We call such a solution u_j^t a ‘traveling wave’ and define a ‘speed’ of this wave by k . Since we see from eq. (1) that the position of the left most 1 of a state must increase as time proceeds, k must be positive.

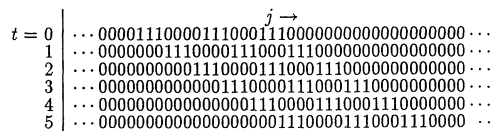
There exist two types of traveling wave solutions. The first type is one which consists of only one sequence of an arbitrary finite number of 1’s. The speed of this solution is equal to the number of 1’s. For example, Fig. 2(a) shows a sequence of three 1’s moving to the right at a speed of 3. We call this type of solution a ‘simple solitary wave’ and the number of 1’s a ‘length’ of the wave. The second type is one which includes an arbitrary finite number of simple solitary waves of the same length and apart from one another at a distance equal to or greater than the length. The speed of this solution is equal to the length of the simple solitary waves included. We call this type of traveling wave solution a ‘compound solitary wave’. Figure 2(b) shows an example of time evolution of this type of

solution.

We now consider an interaction between two solitary waves. For simplicity, let us assume that both waves are simple solitary waves. For example, Fig. 3(a) shows such an interaction between two simple solitary waves, one of which is 3 in length and the other of which is 1. At $t=0$, the longer is located on the far left of the shorter. As time proceeds, the longer approaches the shorter because the longer moves faster. At $t=5$, they exchange the positions. Then they never interact and move to the right at their own speed. Figure 3(b) shows an interaction of the same simple solitary waves the initial positions of which are different from those of Fig. 3(a). In this case, they become two sequences of two 1’s at $t=5$ when the longer overtakes the shorter. However, they also regain their profile after



(a)



(b)

Fig. 2. Solitary wave solutions: (a) simple solitary wave, (b) compound solitary wave.

	$j \rightarrow$
state	$\cdots 0\ 0\ 0\ 0\ 1\ 1\ 1\ 0\ 0\ 1\ 1\ 1\ 0\ 0\ 0\ 1\ 0\ 0\ 0\ 0\ 1\ 1\ 0\ 0\ 0\ 0\ 0\ 0\ \cdots$
step (i)	$\cdots 0_1\ 0_1\ 0_1\ 0_1\ 1_2\ 1_2\ 1_2\ 0_3\ 0_3\ 1_4\ 1_4\ 1_4\ 0_5\ 0_5\ 0_5\ 1_6\ 0_7\ 0_7\ 0_7\ 1_8\ 1_8\ 0_9\ 0_9\ 0_9\ 0_9\ \cdots$
step (ii), (iii)	$\cdots 0_1\ 0_1\ 0_1\ 0_1\ 1_2\ 1_2\ 1_2\ 0_3^2\ 0_3^2\ 1_4^2\ 1_4^2\ 1_4\ 0_5\ 0_5\ 0_5\ 1_6\ 0_7\ 0_7\ 0_7\ 1_8\ 1_8\ 0_9\ 0_9\ 0_9\ 0_9\ \cdots$
step (iv), (i)	$\cdots 0_1\ 0_1\ 0_1\ 0_1\ 1_2\ 1_2\ 1_2\ 0^2\ 0^2\ 1^2\ 1^2\ 1_2\ 0_3\ 0_3\ 0_3\ 1_4\ 0_5\ 0_5\ 0_5\ 1_6\ 1_6\ 0_7\ 0_7\ 0_7\ 0_7\ \cdots$
step (ii), (iii)	$\cdots 0_1\ 0_1\ 0_1\ 0_1\ 1_2\ 1_2\ 1_2\ 0^2\ 0^2\ 1^2\ 1^2\ 1_2\ 0_3\ 0_3\ 0_3\ 1_4^1\ 0_5^1\ 0_5\ 0_5\ 1_6\ 1_6\ 0_7\ 0_7\ 0_7\ 0_7\ \cdots$
step (iv), (i)	$\cdots 0_1\ 0_1\ 0_1\ 0_1\ 1_2\ 1_2\ 1_2\ 0^2\ 0^2\ 1^2\ 1^2\ 1_2\ 0_3\ 0_3\ 0_3\ 1^1\ 0^1\ 0_3\ 0_3\ 1_4\ 1_4\ 0_5\ 0_5\ 0_5\ 0_5\ \cdots$
step (ii), (iii)	$\cdots 0_1\ 0_1\ 0_1\ 0_1\ 1_2^1\ 1_2^1\ 1_2^1\ 0^2\ 0^2\ 1^2\ 1^2\ 1_2^1\ 0_3^1\ 0_3^1\ 0_3^1\ 1^1\ 0^1\ 0_3^1\ 0_3\ 1_4\ 1_4\ 0_5\ 0_5\ 0_5\ 0_5\ \cdots$
step (iv), (i)	$\cdots 0_1\ 0_1\ 0_1\ 0_1\ 1^4\ 1^4\ 1^4\ 0^2\ 0^2\ 1^2\ 1^2\ 1^4\ 0^4\ 0^4\ 0^4\ 1^1\ 0^1\ 0^4\ 0_1\ 1_2\ 1_2\ 0_3\ 0_3\ 0_3\ 0_3\ \cdots$
step (ii), (iii)	$\cdots 0_1\ 0_1\ 0_1\ 0_1\ 1^4\ 1^4\ 1^4\ 0^2\ 0^2\ 1^2\ 1^2\ 1^4\ 0^4\ 0^4\ 0^4\ 1^1\ 0^1\ 0^4\ 0_1\ 1_2^1\ 1_2^1\ 0_3^1\ 0_3\ 0_3\ 0_3\ \cdots$
step (iv), end	$\cdots 0\ 0\ 0\ 0\ 1^4\ 1^4\ 1^4\ 0^2\ 0^2\ 1^2\ 1^2\ 1^4\ 0^4\ 0^4\ 0^4\ 1^1\ 0^1\ 0^4\ 0\ 1^2\ 1^2\ 0^2\ 0\ 0\ 0\ 0\ \cdots$

Fig. 4. Procedures to calculate BS's from a state. Subscripts of elements denote the subset number and superscripts denote the element number.

$t =$	$j \rightarrow$
0	$\cdots 0\ 0\ 0\ 0\ 0\ 1^3\ 1^3\ 1^3\ 0^3\ 0^3\ 0^3\ 0\ 1^3\ 1^3\ 1^3\ 0^3\ 0^3\ 1^3\ 1^3\ 1^3\ 0^3\ 0^3\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ \cdots$
1	$\cdots 0\ 0\ 0\ 0\ 0\ 0\ 0\ 1^3\ 1^3\ 1^3\ 0^3\ 0^3\ 0\ 1^3\ 1^3\ 1^3\ 0^3\ 0^3\ 1^3\ 1^3\ 1^3\ 0^3\ 0^3\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ \cdots$
2	$\cdots 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 1^3\ 1^3\ 1^3\ 0^3\ 0^3\ 0\ 1^3\ 1^3\ 1^3\ 0^3\ 0^3\ 1^3\ 1^3\ 1^3\ 0^3\ 0^3\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ \cdots$
3	$\cdots 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 1^3\ 1^3\ 1^3\ 0^3\ 0^3\ 0\ 1^3\ 1^3\ 1^3\ 0^3\ 0^3\ 1^3\ 1^3\ 1^3\ 0^3\ 0^3\ 0\ 0\ 0\ 0\ 0\ 0\ \cdots$
4	$\cdots 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 1^3\ 1^3\ 1^3\ 0^3\ 0^3\ 0\ 1^3\ 1^3\ 1^3\ 0^3\ 0^3\ 1^3\ 1^3\ 1^3\ 0^3\ 0^3\ 0\ 0\ 0\ 0\ \cdots$
5	$\cdots 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 0\ 1^3\ 1^3\ 1^3\ 0^3\ 0^3\ 0\ 1^3\ 1^3\ 1^3\ 0^3\ 0^3\ 1^3\ 1^3\ 1^3\ 0^3\ 0^3\ 0\ 0\ \cdots$

Fig. 5. Expression in terms of BS's for the time evolution given in Fig. 2(b). Superscripts denote the element number.

BS's are completely separated each other till $t=n$. At $t=n$, the distance between the BS's, $x_2^n - x_1^n$, satisfies $2l_1 \leq x_2^n - x_1^n < 3l_1 - l_2$ and they begin to interact at the next time. There are two cases for the interaction. The first is the case of $2l_1 = x_2^n - x_1^n$. In this case, at $t=n+1$, all 0's of the longer BS appear on the right of the shorter and the shorter becomes the 'complement' of the state at $t=n$. Here, a complement of a BS $\{u_j\}$ is defined by the BS $\{u'_j | u'_j = 1 - u_j\}$. At $t=n+2$, the whole of the longer BS appears on the right of the shorter and the shorter again becomes the complement of the state at $t=n+1$. At $t>n+2$, they are separated each other and move to the right as if each of them exists alone. The second is the case of $2l_1 < x_2^n - x_1^n < 3l_1 - l_2$. At $t=n+1$, some of 0's of the longer appear on the right of the shorter. At $t=n+2$, all 0's and some 1's of the longer appear on the right of the shorter and the shorter becomes the complement of the state at $t=n+1$. At $t=n+3$, the whole of the longer appear on the right to the shorter, the shorter becomes the complement of the state at $t=n+2$, and both BS's are completely separated each other. At $t>n+3$, they move

to the right as if each of them exists alone. If we define a 'phase shift' of BS by the shift of its position due to the interaction with another wave, the phase shift of the longer and the shorter is $+2l_1$ and $-2l_1$, respectively, in both cases. Figure 6(a) shows an example of the evolution of two BS's in the former case and Fig. 6(b) shows that in the latter. Evolutions in Figs. 6(a) and 6(b) are the same as those in Figs. 3(a) and 3(b), respectively.

We finally consider the evolution of an arbitrary state in terms of BS's. Though we have not yet obtained a compact algebraic expression for such an evolution, we can prove the following results:

(i) The number of BS's of the same length included in a state do not change during a time evolution. That is, every BS conserves its identity.

(ii) All BS's are separated one another after a long time. Then the longer is located on the right and every BS moves to the right at a speed equal to its length as if it exists alone.

(iii) From the reversibility of this CA and the result (ii), any state may be realized by the evolution from an appropriate state including

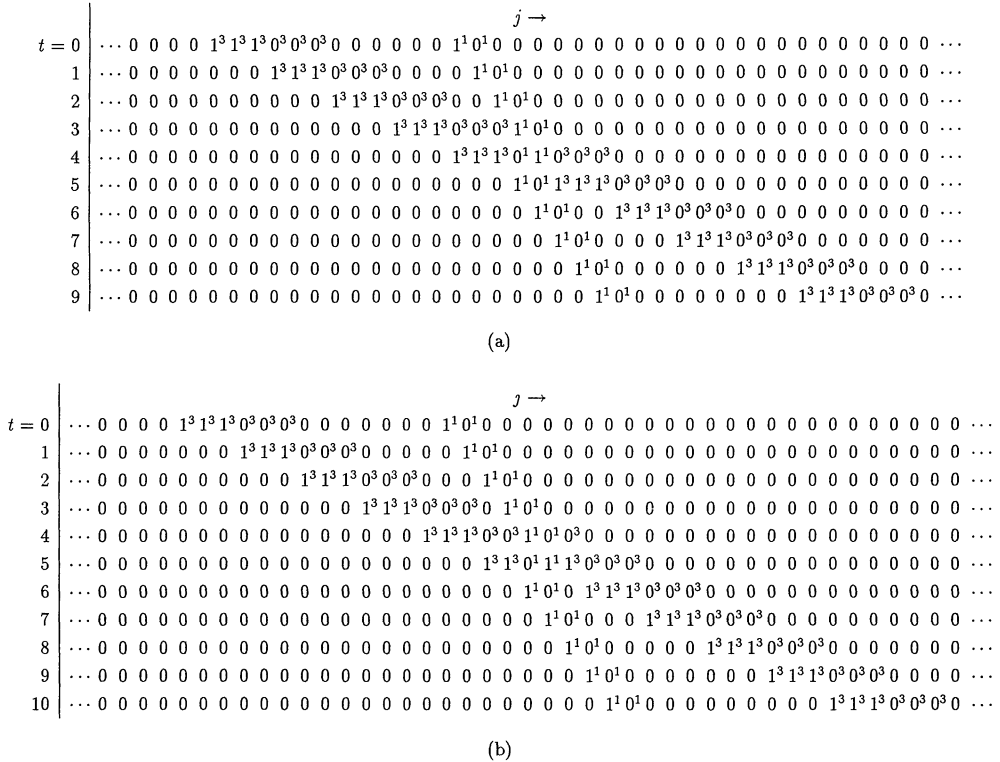


Fig. 6. Interaction between two BS's: (a) and (b) correspond to Figs. 3(a) and 3(b), respectively. Superscripts denote the element number.

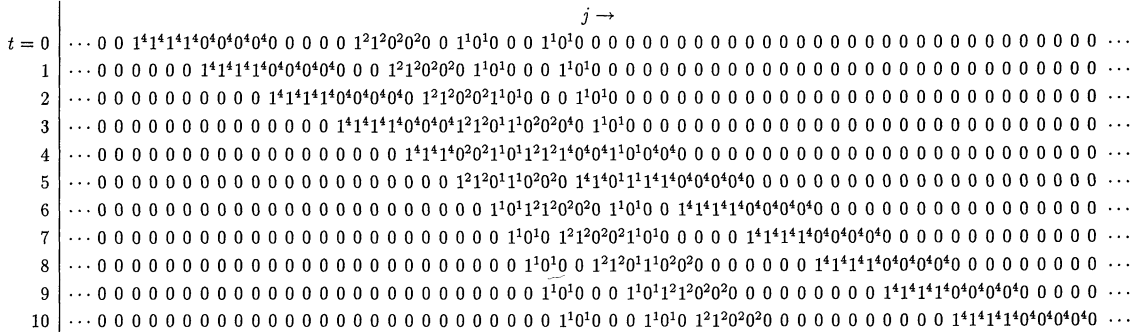


Fig. 7. Expression of evolution from an initial state in terms of BS's. Superscripts denote the element number.

some BS's which are apart enough each other.

(iv) If N denotes the number of all BS's and $l_j (1 \leq j \leq N)$ denotes the length of the j -th BS, respectively, the phase shift s_j of the j -th BS is given by

$$s_j = 2 \left(\sum_{l_i < l_j} l_i - \sum_{l_i > l_j} l_j \right). \quad (7)$$

Figure 7 shows the expression of evolution from an initial state in terms of BS's. From the above results, we see that the behavior of each BS of this CA is analogous to that of solitons in a multi-soliton solution of nonlinear wave equations. For example, the Korteweg-de Vries equation possesses the N -soliton solution as a nonlinear superposition

of N solitons. During the N -soliton solution evolves, each soliton interacts one another preserving their identities and the only effect left with after the interaction is the phase shift.

It is a well known fact that soliton equations

$$\sum_{i=-\infty}^{\infty} u_i^t u_{i+1}^t \sum_{j=1}^{\infty} \prod_{k=1}^j (1 - u_{i+2k}^t) u_{i+2k+1}^t + \sum_{i=-\infty}^{\infty} (1 - u_i^t)(1 - u_{i+1}^t) \sum_{j=1}^{\infty} \prod_{k=1}^j u_{i+2k}^t (1 - u_{i+2k+1}^t). \quad (8)$$

There are another type of conserved quantities. For example, the quantities, $\sum_{i=-\infty}^{\infty} u_i^n$ and $\sum_{i=-\infty}^{\infty} u_i^n u_{i+1}^n$ are conserved. It is noted that the relation between the first and the second type of conserved quantities is not clear yet.

We have proposed a new type of CA and shown that any state of this CA consists of BS's which are considered to be solitons. As for the soliton equations such as the Korteweg-de Vries equation, an initial state generally produce ripples as well as solitons. Since the CA proposed in this paper consists *only* of solitons, it may be one of the simplest soliton systems.

In connection with this, we would touch on an interesting result given in refs. 5 and 6. A class of CA's admits soliton-like excitations,

have a countably infinite number of conserved quantities. This CA also has such quantities. It is seen from the preceding results that the number of BS's equivalent in length is conserved. For example, we can calculate the number of BS's 1 in length at time t by the following formula;

although the system may not be integrable in the sense of soliton equations. The term "integrable on an attractor" has been attached to the system. It may be an interesting problem to study the relationship between the system and the CA we have proposed.

References

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